Yuzhuo Ao

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Yuzhuo Ao | Google Scholar
Hong Kong SAR, China

Research Interest

My research interests primarily focus on machine learning and computer vision. Specifically, I am particularly interested on systems for real-world modeling, representation and interaction.

Education

The Hong Kong University of Science and Technology 2023/9 - 2027/9
BSc in Data Science and Technology with Extended Major in AI (DSCT+AI) *Hong Kong SAR, China*

École Polytechnique Fédérale de Lausanne 2026/2 - 2026/7
Exchange Program in Computer Science section *Lausanne, Switzerland*

Nanyanng Technology University 2025/06 - 2025/07
Summer Exchange Program *Singapore*

Publications

- **SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation** *ECCV 2026*
  Anbang Wang* ,**Yuzhuo Ao***, Shangzhe Wu, Chi-Keung Tang
  ** Equal contribution.*

- **ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation** *arXiv*
  Yuzhuo Ao*, Anbang Wang*, Yu-Wing Tai, Chi-Keung Tang
  ** Equal contribution.*

- **XToM: Exploring the Multilingual Theory of Mind for Large Language Models** *ACL 2026*
  Chunkit Chan, Yauwai Yim, Hongchuan Zeng, Zhiying Zou, Xinyuan Cheng, Zhifan Sun, Zheye Deng, Kawai Chung, **Yuzhuo Ao**, Yixiang Fan, Cheng Jiayang, Ercong Nie, Ginny Y. Wong, Helmut Schmid, Hinrich Schütze, Simon See, Yangqiu Song

Projects

SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation *2025/11 - 2026/03*
Advised by Prof. Chi-Keung Tang and Prof. Shangzhe Wu

- Propose SK-adapter framework, a simple and yet highly efficient and effective framewor that **firstly** unlocks precise skeletal manipulation for native 3D generation. SK-Adapter also allows for precise region-specific local 3D editing based on skeleton prompts.
- Construct the Objaverse-TMS dataset, comprising 24k high-quality text-mesh-skeleton triplets, addressing the data scarcity bottleneck in structure guided 3D generation

ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation *2025/06 - 2025/09*
Advised by Prof. Chi-Keung Tang And Prof. Yu-Wing Tai

- Developed ReasonNavi, a framework that leverages MLLMs for global reasoning and a deterministic planner for local navigation, creating a novel reason-then-act paradigm.
- Enabled unified, zero-shot navigation across diverse tasks (object, image, text goals), eliminating the need for task-specific fine-tuning or reinforcement learning.

Honors and Awards

HKUST Study Abroad Support (2025 Fall)

University's Scholarship Scheme for Continuing Undergraduate Students (2025 Fall, 2024 Fall)

S. S. Chern Class Scholarship (for top math students in HKUST) (2024 Spring)

Dean's List (2025 Fall, 2025 Spring, 2024 Spring, 2023 Fall)

University Admission Scholarship (2023 Summer)

Extracurricular Activity

32nd Executive Committee of Film Society, HKUSTSU 2025/05 - 2026/05
Executive Committee Member *Hong Kong SAR, China*

Skills

Programming Languages	Python (Proficient), C/C++ (Proficient), R (Familiar)
Languages	Mandarin (Native), English (Proficient - IELTS 7.0 [09/2024])